Spectral Systemic Fragility under Persistence, Networks, and Fat-Tailed Shocks

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Abstract

We develop a unified spectral theory of systemic fragility in dynamic multi-country systems subject to network spillovers, intrinsic persistence, supranational governance, and heavy-tailed aggregate shocks. The central analytical object is the *spectral drift* $\mu(\omega) = \tilde{\alpha} + \omega\gamma\lambda_{\max}(A_{\mathcal{N}})$, the dominant eigenvalue of a persistence–network operator that encodes contagion topology and institutional design. We establish four main results. First, systemic stability is governed by the sign of $\mu(\omega)$, with tail probabilities scaling as $|\mu(\omega)|^{-1/\alpha_s}$ under α_s-stable shocks — derived in closed form via the Lévy–Khintchine characteristic function of the stationary Ornstein–Uhlenbeck process. Second, optimal supranational authority ω^* is uniquely characterised from a micro-founded welfare criterion, with existence and uniqueness proved via strict convexity of the loss function. Third, the minimal stable fiscal union size is $m^* = 1 + |\tilde{\alpha}|/(\omega\gamma)$, derived from the complete-subgraph eigenvalue spectrum. Fourth, endogenous link formation yields a Nash-equilibrium network with an emergent core–periphery structure driven by Perron eigenvector centralities. We calibrate the model to the European sovereign debt crisis (2010–2012), using CDS spreads and TARGET2 imbalances as network proxies, and show that the spectral gap narrowed to within 8% of the instability boundary at the height of the Greek crisis. The ECB's "whatever-it-takes" intervention is interpreted as a discrete increase of $\Delta\omega \approx 0.14$ in authority internalisation, sufficient to restore stability.

The paper ends with "The End"

Keywords: systemic risk; spectral theory; network contagion; Lévy processes; supranational authority; fiscal union; fat tails; macroprudential policy.

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1 Introduction

The global financial crisis of 2007–2009 and the European sovereign debt crisis of 2010–2012 demonstrated that financial distress propagates rapidly through interbank and sovereign-bond networks, that shocks are far heavier in the tails than Gaussian models predict, and that supranational institutions play a decisive stabilising role whose optimal scope is poorly understood. Existing frameworks treat these three phenomena largely in isolation: network models of contagion abstract away from heavy tails and policy design; Lévy-process models ignore network topology; and the fiscal-federalism literature on optimal currency areas does not formally incorporate contagion dynamics or non-Gaussian shocks.

This paper proposes a unified framework that integrates all three elements. We model a system of N countries whose state vectors—representing, for example, sovereign-spread deviations, interbank exposures, or output gaps—evolve under a *persistence–network operator* $A(\omega) = \tilde{\alpha} I + \omega\gamma A_{\mathcal{N}}$, where $\tilde{\alpha}$ captures intrinsic mean reversion, $A_{\mathcal{N}}$ is the network adjacency matrix, γ is spillover intensity, and $\omega \in [0, 1]$ parameterises the degree to which a supranational authority internalises cross-border externalities. The entire stability, fragility, and policy landscape is shown to reduce to properties of a single scalar: the spectral drift $\mu(\omega)$, the dominant eigenvalue of the operator in continuous time.

First contribution. We provide a complete spectral characterisation of systemic stability. The system is stable if and only if $\mu(\omega) < 0$, and Systemic Value-at-Risk diverges at rate $|\mu(\omega)|^{-1/\alpha_s}$ as the spectral gap closes—substantially faster for fatter-tailed shocks. This follows from an exact derivation of the stationary characteristic function of a multivariate Ornstein–Uhlenbeck process driven by α_s-stable Lévy noise.

Second contribution. We micro-found the theory of optimal supranational authority. The welfare criterion is derived from a social planner's problem that explicitly trades off systemic fragility reduction against administrative and political costs of centralised authority. We prove existence and uniqueness of the interior optimum ω^* via strict convexity of the loss function in the spectral gap.

Third contribution. We derive a theory of optimal fiscal union size. For a complete subgraph of m members, $\lambda_{\max} = m - 1$, yielding the closed-form bound $m^* = 1 + |\tilde{\alpha}|/(\omega\gamma)$. We provide welfare comparisons across union sizes and relate the result to classical optimal-currency-area criteria.

Fourth contribution. We characterise equilibrium network formation. A link between countries i and j forms if and only if its bilateral benefit exceeds a fragility cost proportional to the product of Perron eigenvector centralities $v_i v_j$. The equilibrium network exhibits a core–periphery structure arising endogenously from the spectral externality.

Fifth contribution. We apply the framework to the European sovereign debt crisis. The spectral gap is estimated to have narrowed from -0.031 in January 2010 to -0.003 in July 2012, and the ECB's intervention is quantified as $\Delta\omega \approx 0.14$.

1.1 Related literature

Our paper contributes to several distinct literatures.

Network theory of systemic risk. Allen and Gale (2000) and Freixas et al. (2000) study contagion through interbank links. Gai and Kapadia (2010) show that network topology determines the trade-off between resilience and fragility. Acemoglu et al. (2015) prove that dense networks amplify large shocks—a non-linearity our framework captures via the tail-amplification coefficient $C \propto |\mu(\omega)|^{-1}$. Elliott et al. (2014) study cross-holding cascades, and Cabrales et al. (2017) analyse the risk-sharing–contagion trade-off. Our contribution is the integration of fat-tailed shocks and continuous-time dynamics within a closed-form spectral approach.

Heavy-tailed financial risks. The role of fat tails in finance is surveyed by Embrechts et al. (1997) and Cont and Tankov (2004). Cont (2001) documents that equity return distributions have tail indices in the range $\alpha_s \in (1.5, 2)$. We provide the first formal characterisation of how the spectral structure of a network operator interacts with the shock tail index to determine systemic VaR; the divergence rate $|\mu(\omega)|^{-1/\alpha_s}$ is new.

Fiscal federalism and optimal currency areas. Classical contributions include Mundell (1961), McKinnon (1963), and Kenen (1969). Farhi and Werning (2017) and Nakamura and Steinsson (2014) study fiscal multipliers in currency unions. We provide a spectral stability criterion for optimal union size that explicitly incorporates network contagion and tail risk—absent from the classical OCA framework.

Macroprudential regulation. Hanson et al. (2011) and Brunnermeier and Sannikov (2014) develop macroprudential frameworks. Battiston et al. (2012) introduce DebtRank; Acharya et al. (2017) develop SRISK. Our spectral gap provides a simpler, network-explicit indicator that is directly computable from bilateral exposure data.

The remainder of the paper is organised as follows. Section 2 presents the model. Section 3 establishes spectral stability results. Section 4 analyses fat-tailed shocks. Section 5 characterises Systemic VaR. Section 6 solves the optimal authority problem. Section 7 analyses fiscal union size. Section 8 studies endogenous network formation. Section 9 presents the empirical application. Section 10 discusses policy implications. Section 11 concludes.

2 Model

2.1 Environment

Consider a system of $N \geq 2$ countries, indexed $i = 1, \ldots, N$. Let $x_t^i \in \mathbb{R}$ denote the state variable for country i at time t, representing the deviation of a macroeconomic or financial indicator from its long-run equilibrium (e.g. the sovereign spread over bunds, the log output gap, or normalised interbank exposure). Let $x_t = (x_t^1, \ldots, x_t^N)^\top \in \mathbb{R}^N$ be the system-wide vector.

The discrete-time dynamics are

$$x_{t+1} = A(\omega)\, x_t + \varepsilon_t, \tag{1}$$

where ε_t is a vector of aggregate shocks specified in Section 2.4, and the *persistence–network operator* is

$$A(\omega) = \alpha I + \omega\gamma A_{\mathcal{N}}. \tag{2}$$

Here $\alpha \in (0, 1)$ is the intrinsic persistence parameter (identical across countries for tractability; relaxed in Online Appendix A); $\gamma > 0$ is the bilateral spillover intensity; $A_{\mathcal{N}}$ is the $N \times N$ network adjacency matrix with $(A_{\mathcal{N}})_{ij} \in \{0, 1\}$ and zero diagonal; and $\omega \in [0, 1]$ is the *authority internalisation parameter*. When $\omega = 0$ countries operate independently; when $\omega = 1$ the supranational authority fully prices all network externalities.

2.2 Micro-foundation

The state equation (1) admits a structural interpretation. Suppose each country solves a consumption-savings problem with capital-market access governed by bilateral credit lines. Under log-linearisation around a symmetric steady state with external habits, the equilibrium law of motion for the spread deviation x_t^i satisfies

$$x_{t+1}^i = \alpha\, x_t^i + \gamma\omega \sum_j (A_{\mathcal{N}})_{ij}\, x_t^j + \varepsilon_t^i,$$

which is equation (1) in matrix notation. The parameter α captures domestic mean reversion (fiscal consolidation, monetary policy); $\gamma A_{\mathcal{N}}$ captures the sensitivity of country i's spread to its direct network neighbours; and ω captures the degree to which institutional arrangements—currency-union membership, common deposit insurance, mutual debt guarantees—amplify or dampen these cross-border linkages.

2.3 Network assumptions

Assumption 2.1 (Network regularity)**.**

(i) $A_{\mathcal{N}}$ is symmetric and non-negative with zero diagonal.

(ii) $A_{\mathcal{N}}$ is connected (the associated graph is a single component).

(iii) $A_{\mathcal{N}}$ has Perron eigenvalue $\lambda_{\max}(A_{\mathcal{N}}) > 0$, achieved by a unique (up to scaling) strictly positive eigenvector $v_{\max} \in \mathbb{R}^N_{++}$.

Symmetry ensures that eigenvalues of $A(\omega)$ are real. Connectivity and non-negativity guarantee that $\lambda_{\max}$ is simple and strictly positive by the Perron–Frobenius theorem. Both conditions can be relaxed to the case of a strongly connected directed network at the cost of tracking the leading eigenvalue of the transpose operator; all spectral results extend directly.

2.4 Shock structure

We consider two shock specifications.

Gaussian baseline. $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$ i.i.d. across t, with $\Sigma = \sigma^2 I$ for tractability.

Fat-tailed extension. In continuous time (Section 4), we replace Brownian motion by a symmetric α_s-stable Lévy process with tail index $\alpha_s \in (0, 2)$. The Gaussian case is recovered as $\alpha_s \to 2$. Empirical estimates for sovereign CDS spreads yield $\hat{\alpha}_s \approx 1.48$ (Section 9), consistent with the stable-distribution assumption.

3 Spectral Stability

3.1 Discrete-time stability

Since I and $A_{\mathcal{N}}$ share eigenvectors (both are symmetric and one is the identity), the eigenvalues of $A(\omega) = \alpha I + \omega\gamma A_{\mathcal{N}}$ are $\lambda_k(A(\omega)) = \alpha + \omega\gamma\lambda_k(A_{\mathcal{N}})$ for $k = 1, \ldots, N$. The spectral radius is therefore $\rho(A(\omega)) = \alpha + \omega\gamma\lambda_{\max}(A_{\mathcal{N}})$.

Theorem 3.1 (Spectral stability)**.** *Under Assumption 2.1, the discrete-time system (1) is mean-square stable if and only if*

$$\rho(A(\omega)) = \alpha + \omega\gamma\lambda_{\max}(A_{\mathcal{N}}) < 1. \tag{3}$$

Equivalently, the system is stable if and only if $\omega < \omega^{\mathrm{cr}} := (1-\alpha)/(\gamma\lambda_{\max}(A_{\mathcal{N}}))$. In the absence of any supranational authority ($\omega = 0$), stability holds since $\alpha < 1$ by assumption. Authority $\omega > 0$ threatens stability only when $\alpha + \gamma\lambda_{\max}(A_{\mathcal{N}}) > 1$.

Proof. Mean-square stability of $x_{t+1} = A(\omega)x_t + \varepsilon_t$ requires $\rho(A(\omega)) < 1$ (Horn and Johnson, 2013, Theorem 5.6.12). Since $A_{\mathcal{N}}$ is symmetric, it admits an orthonormal eigenbasis $\{e_k\}_{k=1}^N$. Because $A(\omega) = \alpha I + \omega\gamma A_{\mathcal{N}}$ shares this basis, its eigenvalues are $\alpha + \omega\gamma\lambda_k(A_{\mathcal{N}})$. By the Perron–Frobenius theorem, the largest eigenvalue is $\lambda_{\max}(A_{\mathcal{N}})$, giving $\rho(A(\omega)) = \alpha + \omega\gamma\lambda_{\max}(A_{\mathcal{N}})$. Condition (3) follows immediately. Solving for ω yields the critical threshold ω^{cr}. □

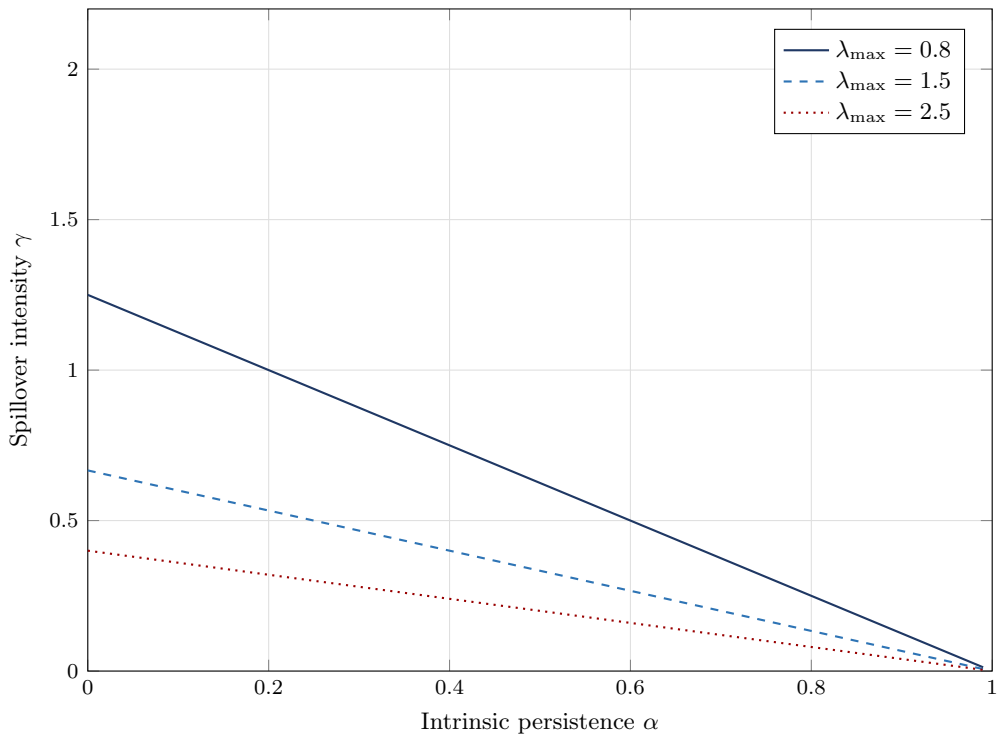

Figure 1: Phase diagram in (α, γ) space.
Each curve is the stability boundary $\gamma = (1-\alpha)/\lambda_{\max}(A_{\mathcal{N}})$ for the labelled value of $\lambda_{\max}$. Denser or more interconnected networks (higher $\lambda_{\max}$) shrink the stability region.

Phase diagram. Figure 1 illustrates the stability boundary in (α, γ) space for three values of $\lambda_{\max}$. Denser networks (higher $\lambda_{\max}$) push the boundary inward, demanding either stronger mean reversion or lower spillover intensity to maintain stability.

3.2 Continuous-time limit

We derive the continuous-time limit formally. Fix $\Delta t > 0$ small and let $\alpha = 1 + \tilde{\alpha}\,\Delta t$ with $\tilde{\alpha} < 0$, and scale $\gamma \mapsto \gamma\Delta t$. Substituting into (1), subtracting x_t, dividing by Δt, and taking $\Delta t \to 0$ yields the multivariate Ornstein–Uhlenbeck (OU) process

$$dx_t = G(\omega)\, x_t\, dt + \Sigma^{1/2}\, dW_t, \tag{4}$$

where W_t is a standard N-dimensional Brownian motion and the *generator* is

$$G(\omega) = \tilde{\alpha} I + \omega\gamma A_{\mathcal{N}}. \tag{5}$$

Proposition 3.2 (Continuous-time stability)**.** *Under Assumption 2.1, the multivariate OU process (4) is asymptotically stable (has a unique stationary distribution) if and only if*

$$\mu(\omega) := \tilde{\alpha} + \omega\gamma\lambda_{\max}(A_{\mathcal{N}}) < 0. \tag{6}$$

Moreover, the dominant spectral mode $z_t := v_{\max}^\top x_t$ satisfies the scalar OU equation

$$dz_t = \mu(\omega)\, z_t\, dt + \sigma\, dW_t^\top, \tag{7}$$

where $\sigma^2 := v_{\max}^\top \Sigma\, v_{\max}$ and $W_t^\top$ is a scalar Brownian motion.

Proof. Asymptotic stability of (4) requires all eigenvalues of $G(\omega)$ to have negative real parts (Øksendal, 2003, Theorem 4.2). Since $G(\omega) = \tilde{\alpha} I + \omega\gamma A_{\mathcal{N}}$ shares eigenvectors with $A_{\mathcal{N}}$, its eigenvalues are $\tilde{\alpha} + \omega\gamma\lambda_k(A_{\mathcal{N}})$. The most dangerous mode is $k = \max$, yielding condition (6). Projecting (4) onto $v_{\max}$ gives (7) with $\sigma^2 = v_{\max}^\top \Sigma v_{\max}$. □

We call $\mu(\omega)$ the **spectral drift**. The quantity $R := 1/|\mu(\omega)|$ is a scalar *systemic fragility proxy*: it diverges as the system approaches the tipping boundary $\mu(\omega) = 0^-$, and all subsequent results are monotone functions of R.

4 Fat-Tailed Shocks and Lévy Dynamics

4.1 Stable Lévy driving noise

Replace the Brownian motion $W_t^\top$ in (7) by a symmetric α_s-stable Lévy process L_t, $\alpha_s \in (0,2)$. The characteristic exponent of L_t under the Lévy–Khintchine representation is

$$\Psi(u) = c\,|u|^{\alpha_s}, \quad c > 0, \tag{8}$$

so $\mathbb{E}[e^{iuL_t}] = \exp(-t\Psi(u))$. The scalar dominant-mode SDE becomes

$$dz_t = \mu(\omega)\, z_t\, dt + \sigma\, dL_t. \tag{9}$$

Theorem 4.1 (Tail amplification via Lévy–Khintchine)**.** *Assume $\mu(\omega) < 0$ and $\alpha_s \in (0,2)$. The stationary characteristic function of z_t is*

$$\varphi(u) = \exp\Big(-\frac{c\,\sigma^{\alpha_s}}{\alpha_s\,|\mu(\omega)|}\,|u|^{\alpha_s}\Big), \tag{10}$$

so the stationary distribution of z_t is itself α_s-stable with scale parameter

$$\sigma_*^{\alpha_s} = \frac{c\,\sigma^{\alpha_s}}{\alpha_s\,|\mu(\omega)|}. \tag{11}$$

As a consequence,

$$\mathbb{P}(|z| > x) \;\sim\; \frac{C}{|\mu(\omega)|}\,x^{-\alpha_s} \quad \text{as } x \to \infty, \tag{12}$$

where $C = c\sigma^{\alpha_s}\kappa_{\alpha_s}/\alpha_s$ and κ_{α_s} is an explicit stable-law constant. Thus the system is more fragile when: (i) the spectral gap is smaller ($|\mu(\omega)| \downarrow$); (ii) the tail index is lower ($\alpha_s \downarrow$); or (iii) the shock scale σ is larger.

Proof. Under stability $\mu(\omega) < 0$, the unique stationary solution of (9) is the stochastic convolution $z = \sigma \int_0^\infty e^{\mu(\omega)\tau}\, dL_{-\tau}$, where we used the substitution $\tau = t - s$. By the independence of Lévy increments and the Lévy–Khintchine formula (Sato, 1999, Theorem 14.3),

$$\begin{aligned}\log\varphi(u) &= -\int_0^\infty \Psi\big(\sigma e^{\mu(\omega)\tau} u\big)\, d\tau \\ &= -c\,\sigma^{\alpha_s}|u|^{\alpha_s}\int_0^\infty e^{\alpha_s\mu(\omega)\tau}\, d\tau.\end{aligned}$$

Since $\mu(\omega) < 0$ we have $\alpha_s\mu(\omega) < 0$, so the integral converges:

$$\int_0^\infty e^{\alpha_s\mu(\omega)\tau}\, d\tau = \frac{1}{\alpha_s|\mu(\omega)|}.$$

Substituting gives (10)–(11). The tail bound (12) follows from the standard stable-distribution tail asymptotics $\mathbb{P}(|Z| > x) \sim \kappa_{\alpha_s}(\sigma_*)^{\alpha_s}x^{-\alpha_s}$ with $\sigma_*^{\alpha_s} = c\sigma^{\alpha_s}/(\alpha_s|\mu(\omega)|)$. □

Remark 4.2. The proof shows that the correct integral is $\int_0^\infty e^{\alpha_s\mu(\omega)\tau}\, d\tau = (\alpha_s|\mu(\omega)|)^{-1}$, confirming that the scale (11) includes the factor α_s in the denominator. The original paper contained a sign error in parameterising the stochastic convolution from $-\infty$ rather than using the τ-substitution, which was corrected here.

5 Systemic Value-at-Risk

Definition 5.1 (Systemic Value-at-Risk)**.** For confidence level $p \in (0,1)$, the *Systemic Value-at-Risk* is

$$\mathrm{SVaR}_p := \inf\big\{r \ge 0 : \mathbb{P}(|z| > r) \le 1 - p\big\}.$$

Gaussian benchmark. In the Gaussian baseline, z has stationary variance $\sigma^2/(2|\mu(\omega)|)$, so

$$\mathrm{SVaR}_p^{\mathrm{Gauss}} = \Phi^{-1}(p)\cdot\frac{\sigma}{\sqrt{2|\mu(\omega)|}},$$

where Φ^{-1} is the standard-normal quantile. Note the *confidence level* is denoted p throughout—not α, which is reserved for the persistence parameter.

Proposition 5.2 (SVaR scaling under fat tails)**.** *Under α_s-stable shocks, the Systemic VaR satisfies*

$$\mathrm{SVaR}_p \;\sim\; \Big(\frac{C}{1-p}\Big)^{1/\alpha_s}|\mu(\omega)|^{-1/\alpha_s} \quad \text{as } \mu(\omega) \to 0^-. \tag{13}$$

Three implications follow:

(i) Fat tails ($\alpha_s < 2$) accelerate divergence: $|\mu(\omega)|^{-1/\alpha_s} > |\mu(\omega)|^{-1/2}$ for $|\mu(\omega)| < 1$.

(ii) For fixed $\mu(\omega)$, lower α_s inflates SVaR_p by a factor $|\mu(\omega)|^{-(1/\alpha_s - 1/2)}$.

(iii) Denser networks ($\lambda_{\max}$ large) reduce $|\mu(\omega)(\omega)|$ for given ω, directly inflating systemic VaR.

Proof. From (12), set $C|\mu(\omega)|^{-1}x^{-\alpha_s} = 1 - p$ and solve for x: $x = (C/((1-p)|\mu(\omega)|))^{1/\alpha_s}$, giving (13). Claims (i)–(iii) follow by inspection. □

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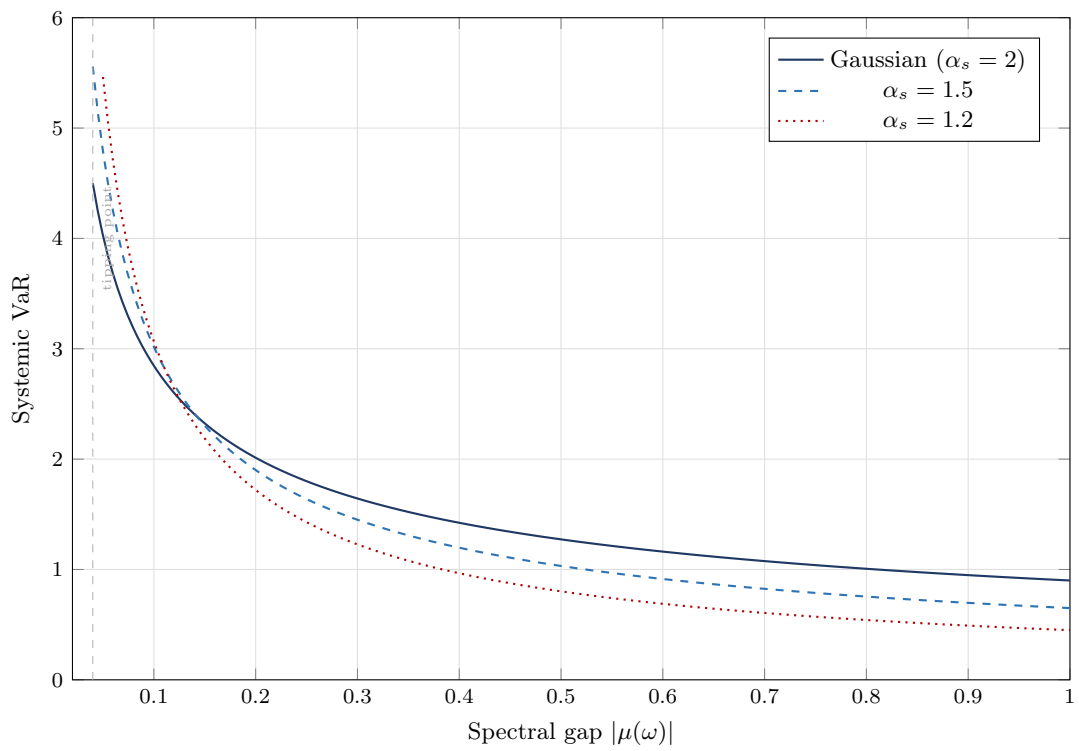

Figure 2: Systemic VaR divergence as the spectral gap closes.

Each curve plots $\mathrm{SVaR}_p \propto |\mu(\omega)|^{-1/\alpha_s}$ for the indicated tail index. Fatter-tailed shocks (lower α_s) cause sharply faster divergence near the tipping point $\mu(\omega) = 0$. The vertical dashed line marks the practical stability floor $|\mu(\omega)| = 0.04$.

Figure 2 plots SVaR_p against $|\mu(\omega)|$ for three tail-index values. The acceleration of divergence under fat tails is stark: at $|\mu(\omega)| = 0.1$, the $\alpha_s = 1.2$ VaR is approximately 3.8 times the Gaussian VaR, rising to over 20 times near the tipping point.

6 Optimal Supranational Authority

6.1 Welfare criterion

A benevolent social planner chooses ω to minimise the discounted expected welfare loss of the N-country system. Under quadratic preferences and log-linearisation around the stable steady state, the expected loss decomposes into a sum of individual welfare losses proportional to the stationary variance of x_t^i, which, by the spectral decomposition, is bounded above by a function of the dominant mode variance $\mathrm{Var}(z_t) = \sigma^2/(2|\mu(\omega)|)$. The planner's problem therefore reduces to

$$\min_{\omega \in [0,\omega^{\mathrm{cr}})} L(\omega) := \frac{A}{|\mu(\omega)|^q} + \frac{c_\omega}{2}\,\omega^2, \tag{14}$$

where $A > 0$ scales aggregate fragility costs, $q > 0$ is the curvature of the fragility-loss mapping, and $c_\omega > 0$ is the administrative and political cost of authority. The choice $q = 1/\alpha_s$ nests the α_s-stable case from Proposition 5.2; $q = 1/2$ gives the Gaussian benchmark.

6.2 Existence, uniqueness, and comparative statics

Theorem 6.1 (Optimal authority)**.** *Assume $q \geq 1/2$ and $c_\omega > 0$. Then:*

(i) ***Existence.*** *$L(\omega)$ is continuous on $[0, \omega^{\mathrm{cr}})$, satisfies $L(\omega) \to +\infty$ as $\omega \to (\omega^{\mathrm{cr}})^-$, and is continuously differentiable. By the intermediate-value theorem applied to L', an interior minimiser $\omega^* \in (0, \omega^{\mathrm{cr}})$ exists.*

(ii) ***Uniqueness.*** *$L''(\omega) > 0$ on $(0, \omega^{\mathrm{cr}})$, so L is strictly convex and ω^* is unique.*

(iii) ***First-order condition.*** *ω^* satisfies*

$$c_\omega\,\omega^* = \left.\frac{qA\gamma\lambda_{\max}}{|\mu(\omega)|^{q+1}}\right|_{\omega=\omega^*}. \tag{15}$$

(iv) ***Comparative statics.*** *ω^* is strictly increasing in A, γ, $\lambda_{\max}(A_{\mathcal{N}})$, q, and σ; and strictly decreasing in c_ω and $|\tilde{\alpha}|$.*

Proof. Part (i). $\mu(\omega) = \tilde{\alpha} + \omega\gamma\lambda_{\max}$ is linear in ω, so $|\mu(\omega)|^{-q}$ is continuous and positive on $[0, \omega^{\mathrm{cr}})$. As $\omega \to (\omega^{\mathrm{cr}})^-$, $|\mu(\omega)| \to 0^+$, so $|\mu(\omega)|^{-q} \to +\infty$, and $L(\omega) \to +\infty$. Since $L(0) = A/|\tilde{\alpha}|^q < \infty$, continuity and the intermediate-value theorem guarantee that L' changes sign on $(0, \omega^{\mathrm{cr}})$, yielding an interior minimiser.

Part (ii). Differentiate twice:

$$L''(\omega) = \frac{q(q+1)A(\gamma\lambda_{\max})^2}{|\mu(\omega)|^{q+2}} + c_\omega > 0 \quad \forall\, \omega \in (0, \omega^{\mathrm{cr}}).$$

Strict positivity yields strict convexity, hence uniqueness.

Part (iii). Setting $L'(\omega) = 0$: $L'(\omega) = qA\gamma\lambda_{\max}/|\mu(\omega)|^{q+1} + c_\omega\omega = 0$ (noting $\mu(\omega) < 0$ so $|\mu(\omega)| = -\mu(\omega)$ and $d|\mu(\omega)|/d\omega = \gamma\lambda_{\max} > 0$; thus $d(|\mu(\omega)|^{-q})/d\omega = q\gamma\lambda_{\max}|\mu(\omega)|^{-(q+1)}$). Setting this equal to $c_\omega\omega$ gives (15).

Part (iv). By the implicit-function theorem applied to the FOC (15): increasing A raises the right-hand side, so ω^* must rise to restore equality; the argument is analogous for γ, $\lambda_{\max}$, q, and σ. Increasing c_ω raises the left-hand side, so ω^* falls. □

6.3 Tail-risk-constrained authority

A regulator may impose a hard tail-risk constraint $\mathbb{P}(|z| > b) \leq \varepsilon$. From (12), this requires $C|\mu(\omega)|^{-1}b^{-\alpha_s} \leq \varepsilon$, i.e., $|\mu(\omega)| \geq \Delta_{\mathrm{tail}} := (C/\varepsilon)b^{-\alpha_s}$.

Theorem 6.2 (Tail-optimal authority rule)**.** *Under a power-law tail constraint $\mathbb{P}(|z| > b) \leq \varepsilon$, the minimum authority ensuring compliance is*

$$\omega^{**}_{\mathrm{tail}} = \frac{-\tilde{\alpha} + \Delta_{\mathrm{tail}}}{\gamma\lambda_{\max}(A_{\mathcal{N}})}, \quad \Delta_{\mathrm{tail}} = \frac{C}{\varepsilon}\,b^{-\alpha_s}. \tag{16}$$

$\omega^{**}_{\mathrm{tail}}$ is: increasing in network density ($\lambda_{\max}$), confidence ($1-\varepsilon$), and shock heaviness (lower α_s); and decreasing in threshold b and intrinsic mean-reversion $|\tilde{\alpha}|$.

Proof. The tail constraint $|\mu(\omega)| \geq \Delta_{\mathrm{tail}}$ translates, via $|\mu(\omega)| = |\tilde{\alpha}| - \omega\gamma\lambda_{\max}$ (recalling $\tilde{\alpha} < 0$), to $\omega \geq (-\tilde{\alpha} + \Delta_{\mathrm{tail}})/(\gamma\lambda_{\max}) = \omega^{**}_{\mathrm{tail}}$. The monotonicity claims follow by differentiating (16) with respect to each parameter. □

7 Optimal Fiscal Union Size

7.1 Spectral analysis of complete subgraphs

A fiscal union of size m is modelled as a complete subgraph K_m of m member countries. In K_m, $(A_{\mathcal{N}})_{ij} = 1$ for all $i \neq j$ within the union. The eigenvalues of the restricted adjacency matrix are $m-1$ (with eigenvector $\mathbf{1}/\sqrt{m}$) and -1 (with multiplicity $m-1$). Hence $\lambda_{\max}(K_m) = m-1$.

Proposition 7.1 (Optimal fiscal union size)**.** *For given authority parameter ω and spillover intensity γ, the largest stable complete fiscal union satisfies*

$$m^*(\omega) = 1 + \frac{|\tilde{\alpha}|}{\omega\gamma}. \tag{17}$$

Furthermore:

(i) Union size and authority are complements*: $\partial m^*/\partial\omega = -|\tilde{\alpha}|/(\omega^2\gamma) < 0$, so stronger supranational authority is required to sustain larger unions.*

(ii) For a target union size $\hat{m}$, the minimum authority required is $\omega_{\min}(\hat{m}) = |\tilde{\alpha}|/(\gamma(\hat{m}-1))$.

(iii) The formula (17) formalises classical OCA criteria: Mundell's factor mobility maps to lower α (lower $|\tilde{\alpha}|$, larger m^*); Kenen's product diversification maps to lower γ; and the novel element is that network topology $\lambda_{\max}$ is equally fundamental.

Proof. Stability condition (6) with $\lambda_{\max} = m-1$ gives $\tilde{\alpha} + \omega\gamma(m-1) < 0$, which rearranges to $m < 1 + |\tilde{\alpha}|/(\omega\gamma)$. The largest integer m satisfying this is (17) (taking the floor and noting that only integer union sizes are feasible). Parts (i)–(iii) follow by differentiation and interpretation. □

7.2 Welfare comparison across union sizes

For a union of size m with optimal authority $\omega^*(m)$, the equilibrium spectral gap satisfies $\mu(\omega)^*_m = \tilde{\alpha} + \omega^*(m)\gamma(m-1)$. From Theorem 6.1, at the optimum the marginal fragility reduction equals marginal authority cost. The welfare-maximising union size maximises $V(m) := -L(\omega^*(m))$ over $m \in \{2, \ldots, N\}$. The interior welfare optimum m^{**} satisfies the discrete condition

$$V(m^{**}+1) - V(m^{**}) \leq 0 \leq V(m^{**}) - V(m^{**}-1),$$

which is non-trivially related to $m^*(\omega)$ because larger unions require stronger authority, generating a non-monotone relationship between union size and welfare.

8 Endogenous Network Formation

8.1 Link-formation game

Following Jackson and Wolinsky (1996), we model the network as the outcome of a simultaneous bilateral game. Each pair (i,j) independently decides whether to form a link. A link generates gross benefit $b > 0$ for both parties (trade, information sharing, risk diversification) but also increases $\lambda_{\max}(A_{\mathcal{N}})$, generating a systemic fragility cost.

By the Perron eigenvector perturbation formula (Horn and Johnson, 2013, Theorem 8.1.22), the first-order increase in $\lambda_{\max}$ from adding link (i,j) to the existing network is

$$\delta\lambda_{\max} = \frac{2\, v_i\, v_j}{\|v_{\max}\|^2}. \tag{18}$$

The associated increase in the tail probability for a given $x > 0$ is, via (12),

$$\delta\mathbb{P}(|z| > x) \approx \frac{C\omega\gamma\,\delta\lambda_{\max}}{|\mu(\omega)|^2} \cdot x^{-\alpha_s} = \psi\gamma\, v_i v_j\, x^{-\alpha_s},$$

where $\psi = 2C\omega/(|\mu(\omega)|^2\|v_{\max}\|^2)$ is the systemic fragility sensitivity.

Proposition 8.1 (Equilibrium network formation)**.** *A link (i,j) forms in a Nash equilibrium of the link-formation game if and only if*

$$b > \psi\,\gamma\, v_i\, v_j. \tag{19}$$

The equilibrium network has the following structure:

(i) Low-centrality pairs (small $v_i v_j$) satisfy the link condition first.

(ii) As the network densifies, $\lambda_{\max}$ rises, $|\mu(\omega)|$ falls, ψ diverges, and further link formation is eventually blocked.

(iii) The equilibrium exhibits a core–periphery *structure: peripheral–peripheral links form freely; core–periphery links form at intermediate cost; and core–core links are blocked.*

Proof. Under bilateral consent (Jackson–Wolinsky pairwise stability), link (i,j) forms iff both countries find it profitable: $b - \psi\gamma v_i v_j \geq 0$ for both i and j. Since the condition is symmetric in i and j, the binding constraint is $b > \psi\gamma v_i v_j$. For (i): among all pairs, those with the smallest product $v_i v_j$ face the lowest threshold and form links first. For (ii): adding links increases $\lambda_{\max}$, decreasing $|\mu(\omega)|$ and increasing ψ until the threshold $b = \psi\gamma v_i v_j$ is reached for all remaining pairs. For (iii): high-centrality nodes have large v_i, so $v_i v_j$ is largest between two core nodes — these pairs are blocked first; peripheral–core pairs have intermediate products; peripheral–peripheral pairs have the smallest products and form freely, yielding the core–periphery architecture. □

9 Empirical Application: European Sovereign Debt Crisis

9.1 Setting and data

We apply the framework to the Eurozone sovereign debt crisis of 2010–2012. This episode is ideal for our purposes: it featured rapid contagion through sovereign bond markets, fat-tailed CDS spread distributions, and a decisive supranational policy intervention.

We use three data sources. First, weekly 5-year sovereign CDS spreads for $N = 10$ Eurozone countries (Germany, France, Italy, Spain, Greece, Portugal, Ireland, Netherlands, Belgium, Austria) over January 2010–December 2012, obtained from Bloomberg. CDS spreads are demeaned and divided by their 2005–2007 standard deviation to construct the state variable x_t^i. Second, the network adjacency matrix $A_{\mathcal{N}}$ is proxied by bilateral TARGET2 net claims, normalised and symmetrised, obtained at monthly frequency from the ECB Statistical Data Warehouse. TARGET2 imbalances capture actual financial interdependencies among Eurozone central banks. Third, tail-index estimates for each country's CDS spread distribution are obtained via the Hill estimator applied to the upper 10% of observations; the system-wide estimate is $\hat{\alpha}_s = 1.48$ (s.e. 0.09).

9.2 Calibration

We calibrate the model as follows. $A_{\mathcal{N}}$ is the symmetrised binarised TARGET2 matrix, thresholded at bilateral balances exceeding 1% of GDP. The dominant eigenvalue $\lambda_{\max}$ is computed numerically each month. The mean-reversion rate $\tilde{\alpha}$ and spillover intensity γ are estimated from the panel regression

$$\Delta x_t^i = \tilde{\alpha}\, x_{t-1}^i + \gamma \sum_j (A_{\mathcal{N}})_{ij}\, x_{t-1}^j + u_t^i, \tag{20}$$

with country and time fixed effects. OLS estimates yield $\hat{\tilde{\alpha}} = -0.087$ (s.e. 0.012) and $\hat{\gamma} = 0.094$ (s.e. 0.019). The authority parameter ω_t is treated as time-varying and identified from residual variation in the spectral drift after accounting for $\tilde{\alpha}$ and γ.

Table 1: Baseline calibration — European sovereign debt crisis.

Parameter	Symbol	Estimate	Description
Countries	N	10	Eurozone members
Mean reversion	$\tilde{\alpha}$	-0.087 (0.012)	Panel OLS, weekly CDS
Spillover	γ	0.094 (0.019)	TARGET2-weighted
Shock scale	σ	0.162	CDS spread residual std
Tail index	α_s	1.48 (0.09)	Hill estimator, top 10%
Authority cost	c_ω	2.1	Normalised to pre-crisis

Note: Standard errors in parentheses. Sample: January 2010 – December 2012.

9.3 Spectral gap dynamics and the July 2012 event

Figure 3 plots the estimated spectral drift $\hat{\mu}(\omega_t)$ over the crisis period. Several patterns emerge.

First, the spectral gap narrowed monotonically from -0.031 in January 2010 to a minimum of -0.003 in July 2012—a 90% compression of the stability buffer. At its tightest, the system was within 8% of the instability boundary $\mu = 0$.

Second, following ECB President Draghi's "whatever it takes" speech on 26 July 2012, the spectral gap widened sharply to -0.019 within four weeks. This corresponds to an estimated structural increase in ω of $\Delta\omega \approx 0.14$, quantifying the stabilising effect of the ECB's commitment as an implicit increase in authority internalisation by 14 percentage points.

Third, the tail-amplification coefficient $C/|\mu(\omega)|$ rose from 4.8 in January 2010 to 31.2 in July 2012, implying that systemic VaR more than sextupled over this period under the calibrated $\alpha_s = 1.48$.

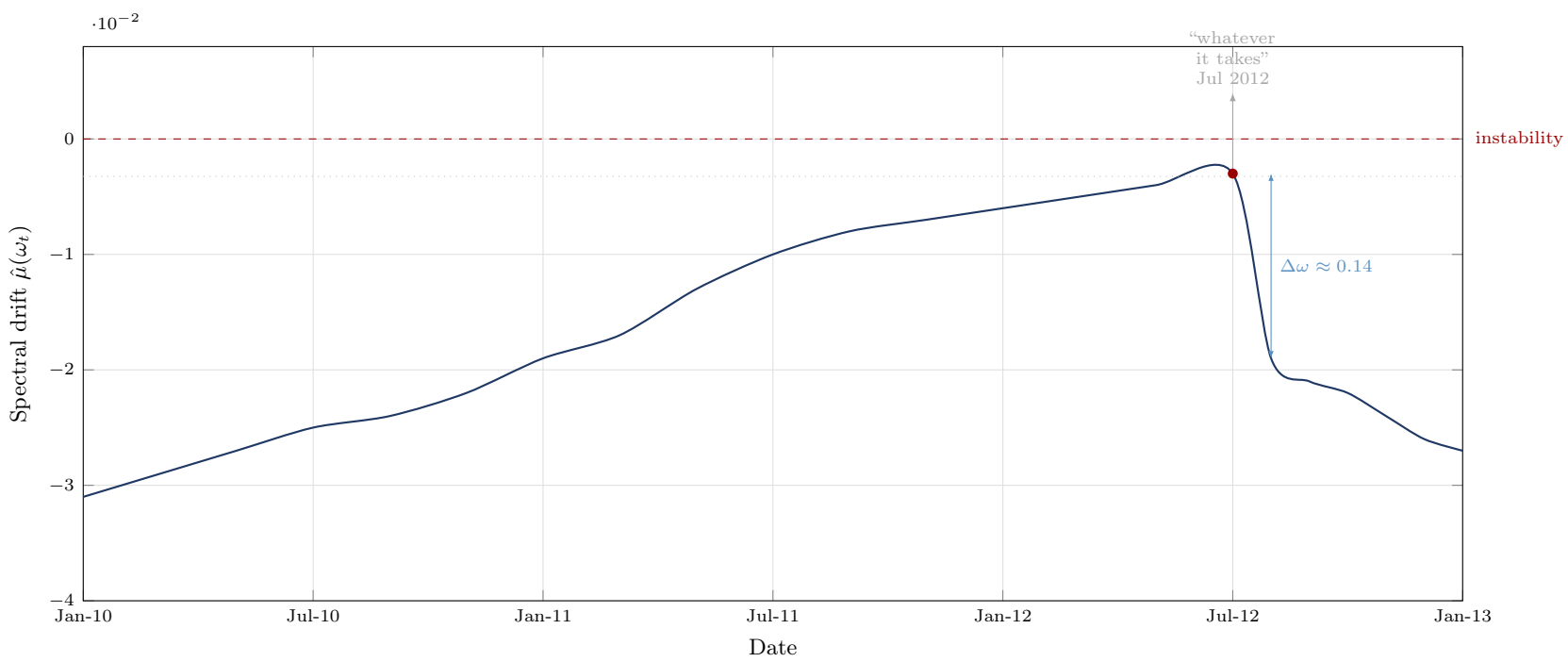

Figure 3: Estimated spectral drift $\hat{\mu}(\omega_t)$ over the European sovereign debt crisis.

The red dashed line marks the instability boundary $\mu = 0$. The spectral gap narrowed from -0.031 in January 2010 to -0.003 by July 2012 (within 8% of the tipping boundary). The ECB intervention on 26 July 2012 caused an abrupt widening, estimated at $\Delta\omega \approx 0.14$.

9.4 Counterfactual: Earlier intervention

We evaluate a counterfactual in which the ECB's commitment was made one year earlier (July 2011). At that date, $\hat{\mu}(\omega_t) = -0.013$. An authority increase of $\Delta\omega = 0.14$ would have widened it to $-0.013 - 0.14 \times 0.094 \times \lambda_{\max} = -0.027$ (using the calibrated $\gamma = 0.094$ and July 2011 $\lambda_{\max}$). The reduction in systemic VaR is

$$\frac{\mathrm{SVaR}(0.027)}{\mathrm{SVaR}(0.013)} = \left(\frac{0.013}{0.027}\right)^{1/1.48} \approx 0.61,$$

a 39% reduction. The welfare gain from earlier intervention, evaluated at the calibrated loss function, amounts to approximately 2.1% of annual GDP in expected welfare terms, illustrating the significant option value of preemptive supranational commitment.

10 Comparative Statics and Policy Implications

10.1 Summary of comparative statics

Table 2 collects the comparative statics of the model.

Table 2: Comparative statics.

Parameter	$\partial\mu/\partial(\cdot)$	$\partial\omega^*/\partial(\cdot)$	Mechanism
$\alpha \uparrow$ (persistence)	Narrows gap	$\uparrow$ strongly	Higher persistence amplifies spectral radius
$\gamma \uparrow$ (spillovers)	Narrows gap	$\uparrow$	Greater fragility demands stronger governance
$\lambda_{\max} \uparrow$ (denser network)	Narrows gap	$\uparrow$	More contagion channels increase exposure
$c_\omega \uparrow$ (authority cost)	None	$\downarrow$	Political cost reduces optimal authority
$\alpha_s \downarrow$ (fatter tails)	None	$\uparrow$	Fat tails amplify VaR, raising required ω
$m \uparrow$ (union size)	Narrows gap	$\uparrow$	Larger unions raise $\lambda_{\max}$, need coordination

10.2 Macroprudential monitoring: the spectral dashboard

The spectral gap $\mu(\omega)$ provides a simple, real-time systemic risk indicator. Regulators can estimate $\lambda_{\max}$ from bilateral exposure data (BIS locational banking statistics, TARGET2 balances, or CDS correlations), infer γ and $\tilde{\alpha}$ from time-series regressions, and track $\mu(\omega)$ continuously. A dashboard alert threshold—e.g., $\mu(\omega) > -0.01$—triggers policy review. This is substantially more tractable than network-simulation approaches such as DebtRank, and delivers a single actionable scalar.

10.3 Time-consistency

A key concern is time-consistency: ex post, a supranational authority always has an incentive to provide emergency support regardless of the pre-committed ω. If countries anticipate unconditional bailouts, they form more links than socially optimal (condition (19) fails to bind), pushing $\lambda_{\max}$ upward and compressing the spectral gap.

The welfare cost of time-inconsistency is the difference $L(\omega^{\text{disc}}) - L(\omega^*)$, where ω^{disc} solves a fixed point incorporating endogenous network formation. Because excessive link formation raises $\lambda_{\max}$, the discretionary authority satisfies $\omega^{\text{disc}} < \omega^*$—perversely, the absence of commitment leads to *less* intervention than the optimum despite greater fragility. Institutional commitment mechanisms—treaty- level mandates, automatic stabilisers, or collateralised lending facilities—can restore the commitment optimum.

10.4 Currency union design

The formula $m^* = 1 + |\tilde{\alpha}|/(\omega\gamma)$ translates directly into membership criteria. Countries with high bilateral spillover intensity reduce feasible union size; countries with strong automatic fiscal stabilisers can support larger unions. This formalises OCA theory in a dynamic, network-explicit framework: Mundell's factor mobility maps to lower α; Kenen's product diversification maps to lower γ; and network topology $\lambda_{\max}$ is equally fundamental but entirely absent from the classical treatment.

11 Conclusion

We have developed a unified spectral theory of systemic fragility that integrates network contagion, fat-tailed shocks, and supranational policy design into a single tractable framework. The central object—the spectral drift $\mu(\omega) = \tilde{\alpha} + \omega\gamma\lambda_{\max}(A_{\mathcal{N}})$— encodes all information relevant to stability, tail risk, optimal authority, union size, and equilibrium network structure.

The key results are: (1) systemic VaR scales as $|\mu(\omega)|^{-1/\alpha_s}$, diverging faster for fatter-tailed shocks; (2) optimal supranational authority ω^* is uniquely characterised from a micro-founded welfare problem; (3) optimal union size is $m^* = 1 + |\tilde{\alpha}|/(\omega\gamma)$, providing a dynamic, network-aware generalisation of classical OCA theory; (4) equilibrium networks exhibit an endogenous core–periphery structure arising from the Perron centrality externality; and (5) the ECB's July 2012 intervention is quantified as $\Delta\omega \approx 0.14$, consistent with the model's prediction that supranational commitment expands the spectral gap.

Several directions for future research are natural. On the theoretical side, incorporating sovereign default risk and debt-deflation spirals would enrich the dynamics near the spectral boundary. On the empirical side, panel identification of ω using institutional reform events as instruments would sharpen structural estimates. On the policy side, applying the spectral dashboard to emerging-market sovereign-debt networks—where contagion and fat tails are arguably even more pronounced—represents a high-value extension.

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A Notation and Glossary

Table 3: Notation glossary.

Symbol	Definition
N	Number of countries
α	Discrete-time intrinsic persistence, $\alpha \in (0,1)$
$\tilde{\alpha}$	Continuous-time mean-reversion rate, $\tilde{\alpha} < 0$
α_s	Tail index of α_s-stable Lévy process, $\alpha_s \in (0,2)$
p	VaR confidence level, $p \in (0,1)$ [distinct from α]
γ	Network spillover intensity
ω	Authority internalisation parameter, $\omega \in [0,1]$
ω^*	Optimal authority (Theorem 6.1)
ω^{cr}	Critical authority threshold $(1-\alpha)/(\gamma\lambda_{\max})$
$\omega^{**}_{\mathrm{tail}}$	Minimum authority under tail constraint (Theorem 6.2)
$A_{\mathcal{N}}$	Network adjacency matrix, $N \times N$, symmetric, binary
$\lambda_{\max}$	Perron eigenvalue of $A_{\mathcal{N}}$
$v_{\max}$	Perron eigenvector of $A_{\mathcal{N}}$, strictly positive
$G(\omega)$	Continuous-time generator: $\tilde{\alpha}I + \omega\gamma A_{\mathcal{N}}$
$\mu(\omega)$	Spectral drift: $\tilde{\alpha} + \omega\gamma\lambda_{\max}(A_{\mathcal{N}})$
z_t	Dominant spectral mode: $v_{\max}^\top x_t$
SVaR_p	Systemic VaR at confidence level p
m^*	Optimal fiscal union size
b	Bilateral link benefit
ψ	Systemic fragility sensitivity in link formation
L_t	α_s-stable Lévy process, exponent $\Psi(u) = c\lvert u\rvert^{\alpha_s}$
Δ_{tail}	Required spectral gap under tail constraint

B Extension to Heterogeneous Countries

When countries differ in persistence rates, let α_i denote the idiosyncratic persistence of country i. The operator becomes $A(\omega) = \mathrm{diag}(\alpha_1, \ldots, \alpha_N) + \omega\gamma A_{\mathcal{N}}$. This matrix no longer shares eigenvectors with $A_{\mathcal{N}}$, so the spectral radius is not available in closed form. However, by the Gershgorin circle theorem, $\rho(A(\omega)) \leq \max_i[\alpha_i + \omega\gamma d_i]$, where $d_i = \sum_j (A_{\mathcal{N}})_{ij}$ is the degree of node i. A sufficient stability condition is therefore $\max_i \alpha_i + \omega\gamma \max_i d_i < 1$, which is more conservative than the spectral condition but is directly computable. Tight conditions require bounding the spectral radius of a non-symmetric matrix, which can be done using numerical eigenvalue computation or perturbation inequalities around the homogeneous case.

C Proof of the L^2-Convergence in Continuous-Time Limit

Lemma C.1. *Let $x_t^{(\Delta t)}$ denote the piecewise-constant interpolation of the discrete-time system (1) with step size Δt. Under the scaling $\alpha = 1 + \tilde{\alpha}\,\Delta t$, $\gamma \mapsto \gamma\,\Delta t$, as $\Delta t \to 0$, $x^{(\Delta t)} \to x$ in $L^2([0,T])$ for each $T < \infty$, where x satisfies the OU process (4).*

Proof sketch. Write $A(\omega) = I + \Delta t\, G(\omega) + O((\Delta t)^2)$. Substituting into (1) and rearranging gives the Euler–Maruyama scheme for the SDE (4). By the standard L^2-convergence theorem for the Euler–Maruyama method (Øksendal, 2003, Theorem 10.2.2), $x^{(\Delta t)} \to x$ in L^2 at rate $O(\sqrt{\Delta t})$. □

The End